

Pós-Graduação em Sistemas Computacionais Embebidos
Sistemas Operativos de Tempo-Real (PGSCE-SIOPTR)

Individual Exercise

Duration: 1h:00m

Version: B

Number: _____

Name: _____

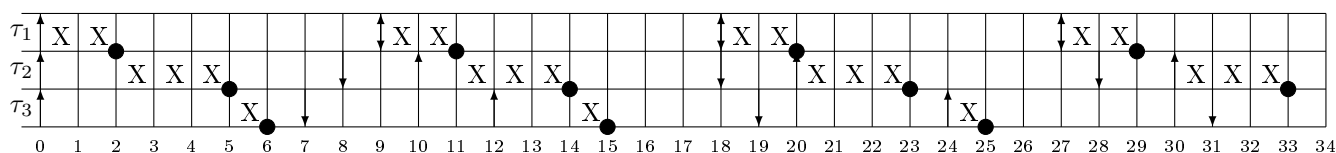
1. Static Priorities(6/20)

Consider the following task set:

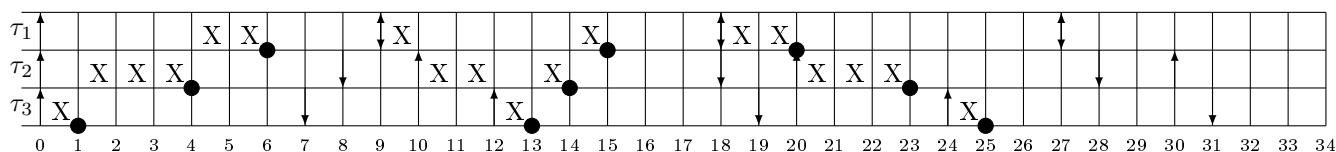
Task	C	T	D
τ_1	2	9	9
τ_2	3	10	8
τ_3	1	12	7

Assume 0 time instant as the activation time instant for all tasks.

- (3/20) – Trace the execution of the task set according to the Rate Monotonic scheduling policy for the first 27 time units.



- (3/20) – Trace the execution of the task set according to the Deadline Monotonic scheduling policy for the first 27 time units.



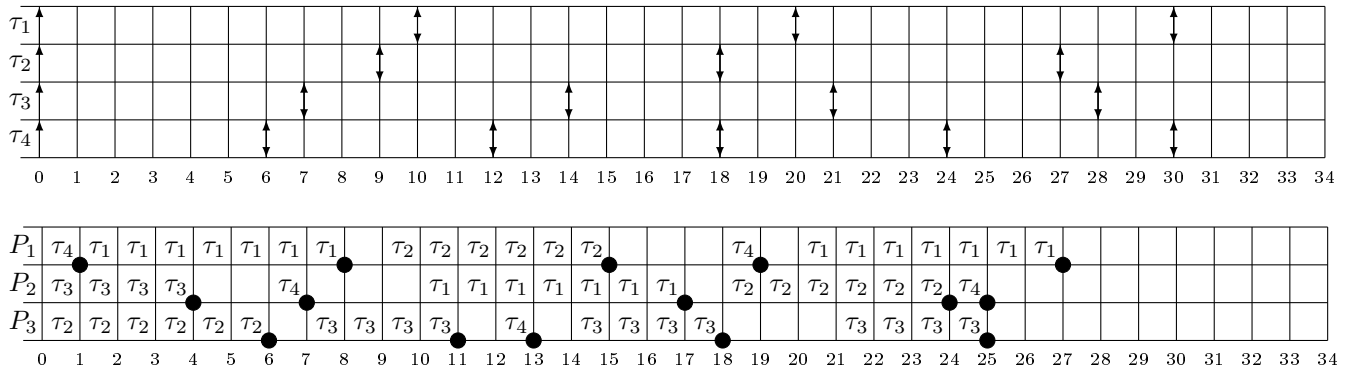
2. Multi-Processor (8/20)

Consider the following task set:

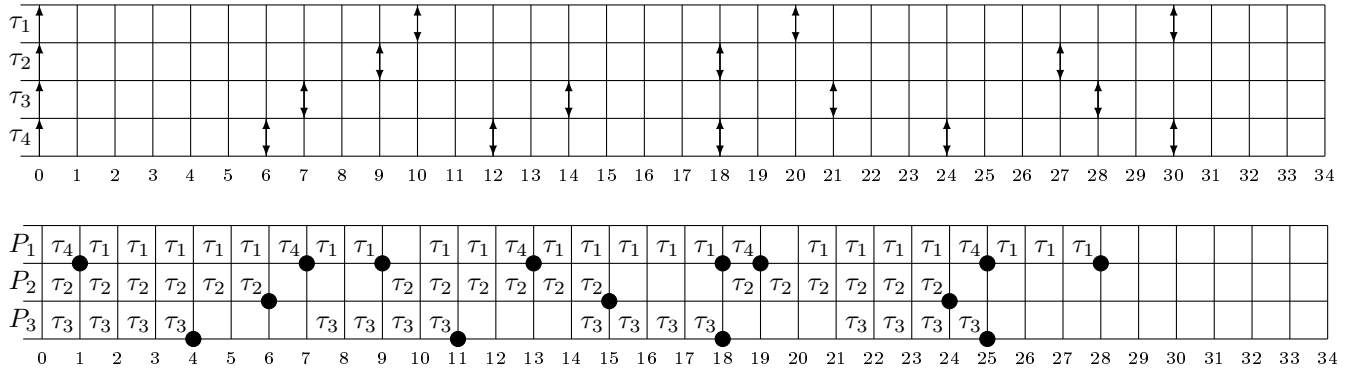
Task	C	T	D	U
τ_1	7	10	10	0.70
τ_2	6	9	9	0.67
τ_3	4	7	7	0.57
τ_4	1	6	6	0.17

Assume 0 time instant as the activation time instant for all tasks and also the computational system is composed by three identical processors.

- (4/20) – Trace the execution of the task set according to the Global EDF (Earliest Deadline First) scheduling policy for the first 28 time units.



- (4/20) – Trace the execution of the task set according to the Partitioned EDF scheduling policy for the first 28 time units.



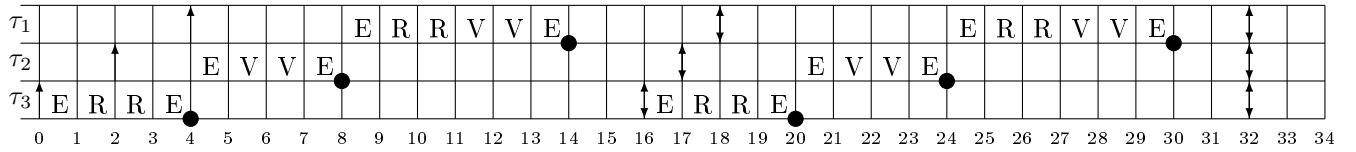
3. Shared Resources (6/20)

Consider the following task set:

Task	C	T = D	Execution sequence	Activation time instant
τ_1	6	14	ERRVVE	4
τ_2	4	15	EVVE	2
τ_3	4	16	ERRE	0

Assume that shared resources are identified in the task execution sequence by letters **V** and **R** and task scheduling policy is according to EDF.

- (3/20) – Trace the execution of the task set according to the Priority Inheritance Protocol (PIP) for the first 30 time units.



- (3/20) – Trace the execution of the task set according to the Immediate Ceiling Priority Protocol (ICPP) for the first 30 time units.

